

MINA SULTAN QABOOS, Oman

WMO: 412580

Lat: 23.628N

Lon: 58.572E

Elev: 3

StdP: 101.29

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	20.0	-0.6	3.6	26.7	3.6	4.9	27.1	8.4	22.9	7.3	22.4	3.1	260	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.0	40.3	22.3	39.2	22.4	38.1	22.6	31.3	32.8	30.6	32.3	29.9	32.0	3.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.0	28.8	32.4	30.2	27.5	31.7	29.5	26.4	31.4	107.8	32.9	103.8	32.3	100.4	31.9	34.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
7.4	6.4	5.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	11.2	31.2	1.2	1.6	10.4	32.3	9.7	33.2	9.0	34.1	8.2	35.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	22.2	23.2	25.8	29.8	33.7	34.0	31.9	30.6	30.3	30.6	27.0	23.9
	DBStd	4.39	1.38	1.71	2.05	2.45	2.43	2.65	2.61	2.44	1.85	1.67	1.42	1.52
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6792	379	369	488	593	735	719	678	639	611	638	511	431
	CDD18.3	3751	121	137	230	343	477	469	419	381	361	379	261	173
	CDH23.3	47466	178	438	1857	4597	7653	7529	6212	5294	4942	5304	2671	788
	CDH26.7	25582	1	52	383	2280	5173	5133	3733	2827	2548	2827	602	23
Wind	WSAvg	2.8	2.7	2.8	2.9	2.8	2.8	2.9	3.0	3.2	2.7	2.4	2.4	2.7
Precipitation	PrecAvg	89	25	27	15	5	0	1	1	5	0	0	4	9
	PrecMax	271	112	104	74	35	6	9	16	86	0	1	37	42
	PrecMin	4	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	65	33	32	21	9	1	2	3	18	0	0	10	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	31.2	33.1	37.4	41.4	41.7	41.0	40.1	37.2	36.8	31.2	28.2
		MCWB	18.3	16.9	18.5	20.7	21.7	22.7	23.4	23.6	21.4	20.8	19.9	19.6
	2%	DB	25.5	27.9	31.3	35.9	39.8	40.5	39.3	37.7	35.5	35.4	30.3	27.2
		MCWB	19.2	19.5	18.6	19.5	21.1	22.3	23.8	24.2	22.9	21.1	21.6	20.1
	5%	DB	24.9	26.5	29.7	34.6	38.7	39.5	37.3	35.3	34.0	34.2	29.6	26.6
		MCWB	19.6	19.4	19.4	19.7	21.4	22.5	24.8	25.4	23.8	21.5	22.2	20.4
	10%	DB	24.3	25.5	28.6	33.4	37.7	38.3	35.3	33.6	32.9	33.2	29.0	26.0
		MCWB	19.3	19.8	19.7	19.9	21.6	23.2	26.6	26.2	25.3	22.5	22.3	20.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.8	26.0	27.8	30.8	32.3	31.9	31.1	30.2	29.5	26.1	24.5
		MCDB	24.5	24.9	26.9	29.6	32.1	33.4	33.4	31.8	31.5	31.0	28.6	26.0
	2%	WB	21.9	23.3	24.9	26.6	30.0	31.7	30.9	30.1	29.2	28.6	25.2	23.6
		MCDB	23.8	24.6	26.6	28.9	32.1	33.1	32.4	31.5	30.8	30.5	28.4	25.8
	5%	WB	21.1	22.6	23.9	25.9	29.1	31.0	30.2	29.4	28.5	27.7	24.4	22.8
		MCDB	24.0	24.5	26.1	28.9	32.0	32.8	31.8	30.9	30.5	30.4	28.3	25.6
	10%	WB	20.4	21.6	23.0	25.1	28.2	30.1	29.7	28.7	28.1	27.0	23.8	21.7
		MCDB	23.8	24.4	26.2	28.9	32.0	32.8	31.6	30.6	30.2	30.5	28.1	25.2
Mean Daily Temperature Range	5% DB	MDBR	2.7	2.8	3.1	3.8	4.5	4.0	3.1	3.1	2.9	3.0	2.5	2.6
		MCDBR	3.1	4.0	5.0	5.4	6.4	6.3	6.1	6.4	5.2	5.2	3.0	3.1
		MCWBR	3.0	4.2	5.0	5.0	5.5	6.6	5.5	5.0	5.5	6.1	3.8	3.4
	5% WB	MCDBR	2.7	2.6	3.4	3.8	4.1	4.2	3.9	3.7	3.2	3.3	2.6	2.6
		MCWBR	2.7	3.6	4.7	5.1	5.7	5.7	3.8	3.7	3.7	4.7	3.5	3.1
Clear-Sky Solar Irradiance	taub	0.422	0.465	0.525	0.572	0.596	0.669	0.740	0.678	0.577	0.478	0.440	0.422	
	taud	2.242	2.089	1.918	1.837	1.788	1.674	1.587	1.686	1.852	2.063	2.198	2.231	
	Ebn at Noon	830	817	786	757	735	678	631	671	733	793	804	812	
	Edh at Noon	125	156	193	213	223	248	271	245	204	157	130	123	
All-Sky Solar Radiation	RadAvg	4.42	5.23	6.09	6.79	7.42	7.23	7.07	6.58	6.21	5.61	4.66	4.19	
	RadStd	0.21	0.23	0.31	0.34	0.15	0.25	0.29	0.18	0.12	0.14	0.14	0.21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.23	N/A	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98

Nomenclature: See separate page